CSC 6302 Practical 5 — Image Segmentation

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Detection draws a box; segmentation answers which pixels belong. I work through global and adaptive thresholding with a hand-made mask and my own IoU, marker-based watershed on touching coins, GrabCut on a portrait object, then DeepLabV3 (semantic) versus Mask R-CNN (instance), and a Gradio segmentation studio.

Task 1 — Thresholding, honestly evaluated

I built an unevenly-lit document page (strong left–right brightness ramp) so a single threshold has to fail somewhere. For a representative crop I painted a reference mask myself (stroke regions + ink darkening), saved as `images/prac5_handmade_mask.png`. IoU is my NumPy function: intersection of binary masks over their union.

IoU results: fixed = 0.1266, otsu = 0.0904, adaptive_mean = 0.1795, adaptive_gaussian = 0.1621.

Adaptive wins when lighting ramps across the page: a single threshold cannot mark dark-side ink and bright-side ink at once. Otsu picks one global cut; fixed is worse if my guess is off. Adaptive windows follow local contrast, so ink stays foreground under the ramp. fixed_threshold=90

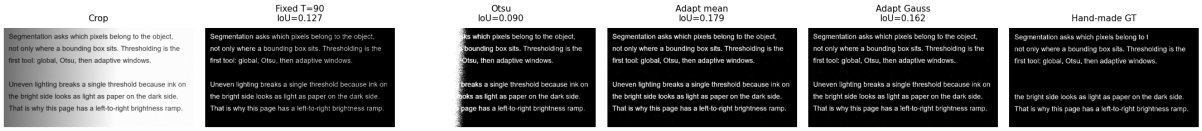


Figure: crop, fixed, Otsu, adaptive mean/Gaussian, hand-made GT.

Task 2 — Watershed for touching objects

Touching coins defeat plain connected components: the binary blob is one component. Marker-based watershed: threshold → morphological opening → distance transform → sure-foreground peaks → `cv2.watershed`. Intermediates are saved under `report/figures/t2_*.png`.

manual_count=15
watershed_markers=16
watershed_labels=16
note=Touching coins merge under connected components; distance peaks give one marker per coin when the peak threshold is tuned. Over-split happens if specular blobs create extra peaks; merges happen when two coins share one distance ridge below the sure-FG cut.

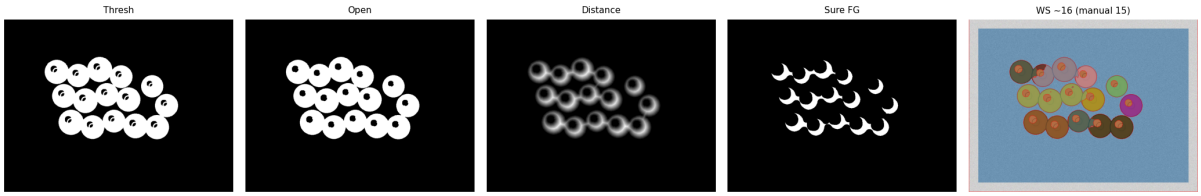


Figure: threshold, opening, distance, sure FG, watershed boundaries.

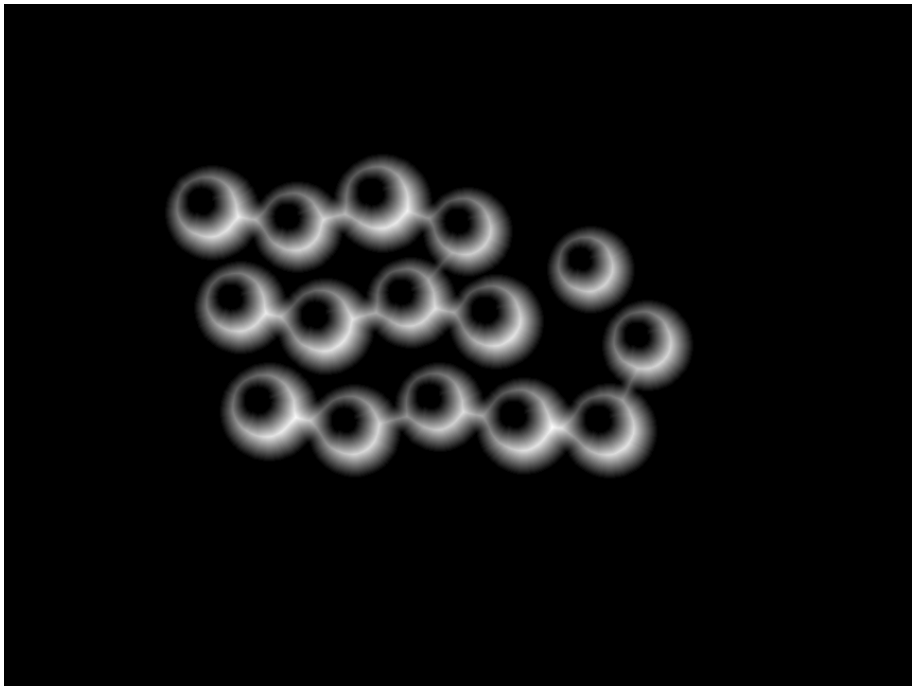


Figure: Distance transform.

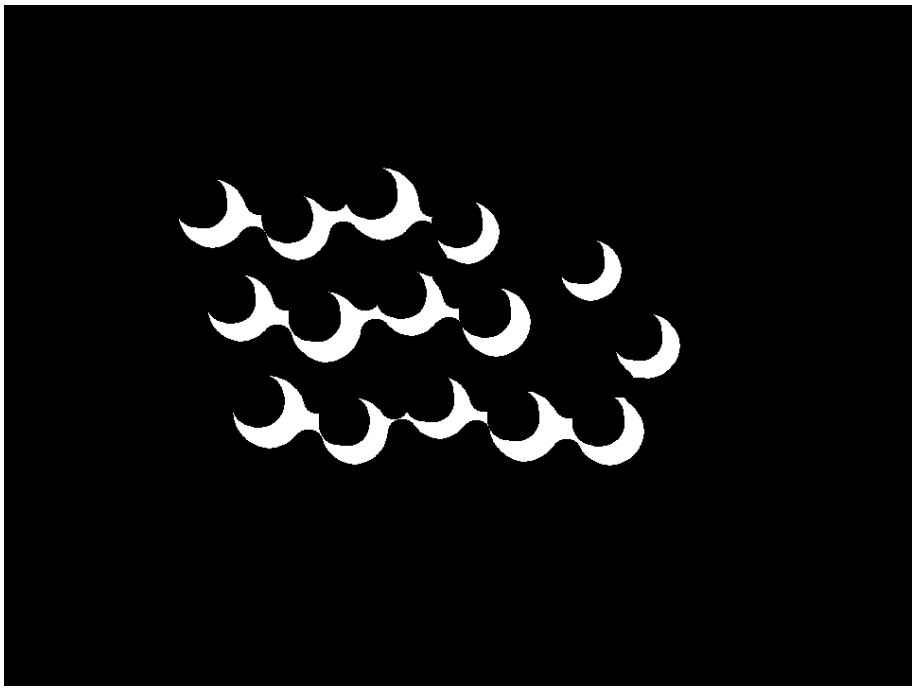


Figure: Sure-foreground markers.

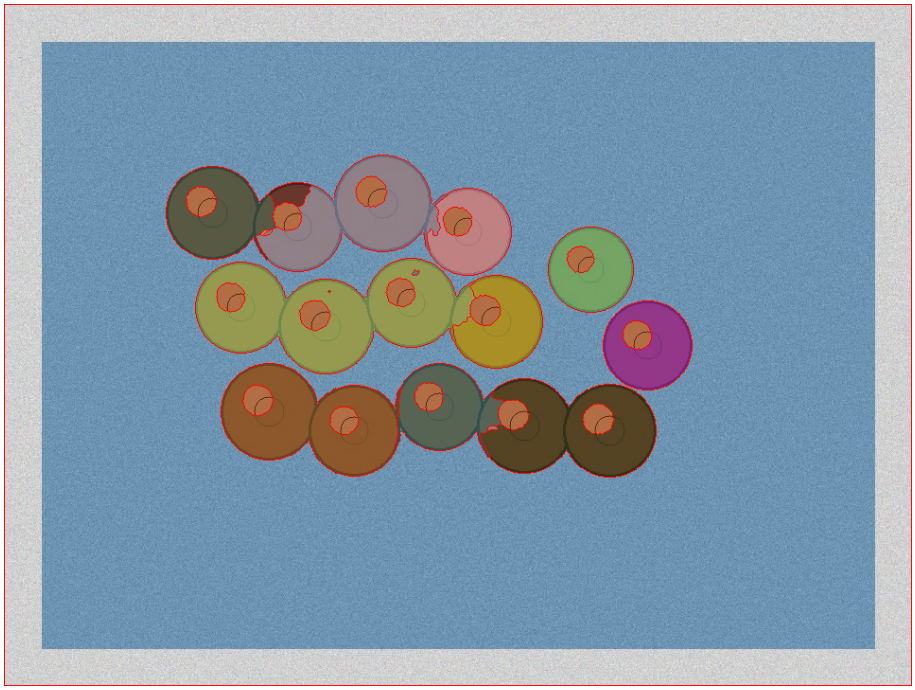


Figure: Final coloured watershed.

Task 3 — GrabCut

I cut the creatine jar (and hand) from `portrait.jpg` with a bounding rectangle, then compared 1 vs 5 iterations, then added definite background strokes at the borders and a definite foreground circle on the label.

GrabCut iterates a Gaussian Mixture Model (GMM) for foreground and background colours, then a graph-cut that labels each pixel. Extra iterations refine the GMMs; brush hints clamp some pixels as hard FG/BG.



Figure: rect seed, 1 iter, 5 iters, mask hints.

Task 4 — Semantic vs instance

I ran torchvision DeepLabV3-ResNet50 (VOC semantic) and Mask R-CNN-ResNet50-FPN (COCO instance) on my street scene and on a group street photo with many people. Semantic segmentation paints every person pixel with one 'person' class; instance segmentation gives each person a separate coloured mask.

deeplab_street_ms=2063.1
maskrcnn_street_ms=7185.3
deeplab_group_ms=5664.9
maskrcnn_group_ms=7445.4
group_instances=24
street_instances=1

DeepLabV3 is an encoder–decoder idea like U-Net: a ResNet encoder builds rich semantics at low resolution; ASPP + decoder upsample back to pixels. U-Net adds skip links from encoder to decoder; DeepLab leans on dilated convolutions / ASPP instead of dense skips, but the contract–expand story is the same lecture idea.



Figure: group photo — one person blob vs separate instances.

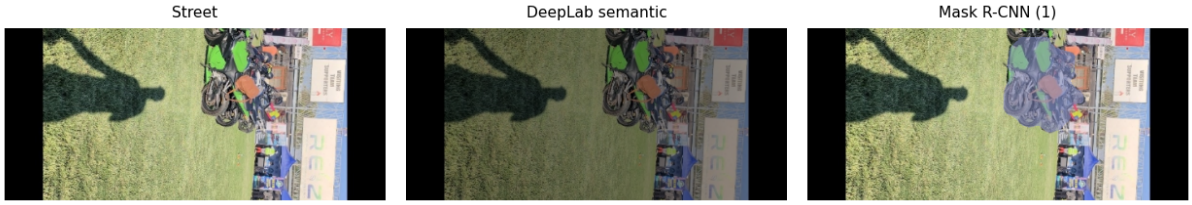


Figure: street scene — DeepLab vs Mask R-CNN.

Failure case

Failure case: my patterned tablecloth (chitenge-like geometry) is not a VOC/COCO class. DeepLab maps busy fabric to 'background' or nearby furniture/plant bits; Mask R-CNN may hallucinate a 'dining table' box but cannot name the print. Models only know labels they were trained on. failure_deeplab_ms=3800.7



Figure: patterned cloth — classes the model was never taught.

Task 5 — Deployed segmentation studio

Gradio app (`prac5/app.py`): image upload; dropdown Otsu / adaptive / watershed / DeepLabV3; overlay output; watershed object count; DeepLab class legend; UI note that DeepLab on CPU takes a few seconds. App source: https://huggingface.co/Salaam21/csc6302-prac5-SINYANI . Creating a live Gradio Space returned HTTP 402 — restricted by Hugging Face PRO (same as Practical 4). Local: `python prac5/app.py`.

Reproducibility & integrity

Reproduce: `pip install -r prac5/requirements.txt`, then `python prac5/prepare_images.py`, `python prac5/run_prac5.py`, `python prac5/app.py`. Composer (Cursor) helped wire OpenCV/torchvision and the Gradio shell; IoU, mask painting choices, watershed discussion, and this report wording are mine. The hand-made mask is my annotation for this crop — not shared with other students.

References: OpenCV docs (threshold, watershed, GrabCut); torchvision DeepLabV3 / Mask R-CNN model cards; course lectures on U-Net and segmentation.